

Exercise 4 - Theoretical Questions

Question 1: Lazy Lists

IDO KOZIH
IDO UZIHAYO

June 2026

Question 1.1: Lazy Lists

(a) Equivalence Criterion

Two lazy lists are considered equivalent if, after any finite number of applications of `tail`, either both lists are empty, or both lists have a defined `head` and these two `head` values are equal.

(b) Equivalence of `even-square-1` and `even-square-2`

Let

$$L_1 = \text{even-square-1}$$

and

$$L_2 = \text{even-square-2}.$$

We consider an arbitrary natural number T produced by `integers-from 0`.

In L_1 , the number T is first mapped to T^2 , and then the filter checks whether T^2 is even.

In L_2 , the number T is first filtered according to whether T is even, and only then it is mapped to T^2 .

We split into two cases.

If T is even, then T^2 is also even. Therefore, in L_1 , the value T^2 passes the filter. In L_2 , the value T passes the filter and is then mapped to T^2 . Thus, in both lists, the same value T^2 appears.

If T is odd, then T^2 is also odd. Therefore, in L_1 , the value T^2 does not pass the filter. In L_2 , the value T does not pass the filter before the mapping. Thus, in both lists, T contributes no value to the final lazy list.

Hence, for every natural number T , either both lists include the same value T^2 , or both lists exclude it. Since both `map` and `filter` preserve the original order of the remaining elements, both lazy lists produce the same sequence:

$$0, 4, 16, 36, 64, \dots$$

Therefore, according to the equivalence criterion defined in part (a), L_1 and L_2 are equivalent lazy lists.